

Exploratory Data Analysis

Slot :- E2 +TE2

Assignment = 1

```
1. import pandas as pd
import numpy as np

# Corrected variable name syntax
dataFrame1 = pd.DataFrame({
    'StudentID': [1, 3, 5, 7, 9],
    'Score': [89, 39, 50, 97, 22]
})
\dataFrame2 = pd.DataFrame({
    'StudentID': [2, 4, 6, 8, 10],
    'Score': [98, 93, 44, 77, 69]
})

# Vertical Concatenation
vertical_concat = pd.concat([dataFrame1, dataFrame2], ignore_index=True)
print("Vertical Concatenation:")
print(vertical_concat)

# Horizontal Concatenation
horizontal_concat = pd.concat([dataFrame1, dataFrame2], axis=1)
print("\nHorizontal Concatenation:")
print(horizontal_concat)
```

... Vertical Concatenation:

	StudentID	Score
0	1	89
1	3	39
2	5	50
3	7	97
4	9	22
5	2	98
6	4	93
7	6	44
8	8	77
9	10	69

Horizontal Concatenation:

	StudentID	Score	StudentID	Score
0	1	89	2	98
1	3	39	4	93
2	5	50	6	44
3	7	97	8	77
4	9	22	10	69

```
2. dfSE = pd.DataFrame({  
    'StudentID': [2, 4, 6, 8, 10, 12],  
    'ScoreSE': [98, 93, 44, 77, 69, 56]  
})
```

```
dfML = pd.DataFrame({  
    'StudentID': [1, 2, 3, 4, 5, 6, 8, 10],  
    'ScoreML': [39, 93, 49, 44, 55, 78, 97, 87]  
})
```

Inner Join

```
inner_join = pd.merge(dfSE, dfML, on='StudentID', how='inner')  
print("\nInner Join:")  
print(inner_join)
```

Left Join

```
left_join = pd.merge(dfSE, dfML, on='StudentID', how='left')  
print("\nLeft Join:")  
print(left_join)
```

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Right Join

```
right_join = pd.merge(dfSE, dfML, on='StudentID', how='right')
```

```
print("\nRight Join:")
```

```
print(right_join)
```

Outer Join

```
outer_join = pd.merge(dfSE, dfML, on='StudentID', how='outer')
```

```
print("\nOuter Join:")
```

```
print(outer_join)
```

```
Inner Join:
***
  StudentID  ScoreSE  ScoreML
0         2        98        93
1         4        93        44
2         6        44        78
3         8        77        97
4        10        69        87

Left Join:
  StudentID  ScoreSE  ScoreML
0         2        98      93.0
1         4        93      44.0
2         6        44      78.0
3         8        77      97.0
4        10        69      87.0
5        12         56       NaN

Right Join:
  StudentID  ScoreSE  ScoreML
0         1        NaN       39
1         2      98.0       93
2         3        NaN       49
3         4      93.0       44
4         5        NaN       55
5         6      44.0       78
6         8      77.0       97
7        10      69.0       87

Outer Join:
  StudentID  ScoreSE  ScoreML
0         1        NaN      39.0
1         2      98.0      93.0
2         3        NaN      49.0
3         4      93.0      44.0
4         5        NaN      55.0
5         6      44.0      78.0
6         8      77.0      97.0
7        10      69.0      87.0
8        12      56.0       NaN
```

```
3. sales = pd.DataFrame({
    'Quantity': [100, 250, 300, 150],
    'UnitPrice': [50, 60, 45, 80]
})
```

```
sales['TotalPrice'] = sales['Quantity'] * sales['UnitPrice']
```

```
print("\nSales Data:")
```

```
print(sales)
```

```
...
Sales Data:
   Quantity  UnitPrice  TotalPrice
0        100         50        5000
1        250         60       15000
2        300         45       13500
3        150         80       12000
```

```
4. print("\nDescriptive Statistics:")
```

```
print(sales.describe())
```

```
...
Descriptive Statistics:
   Quantity  UnitPrice  TotalPrice
count      4.000000      4.00000      4.000000
mean     200.000000     58.75000    11375.000000
std       91.287093     15.47848     4422.951503
min      100.000000     45.00000     5000.000000
25%      137.500000     48.75000    10250.000000
50%      200.000000     55.00000    12750.000000
75%      262.500000     65.00000    13875.000000
max      300.000000     80.00000    15000.000000
```

```
5. import numpy as np
```

```
import pandas as pd
```

```
company_df = pd.DataFrame({"Company": ["ABC", "XYZ", "ABC", "PQR", "XYZ", "ABC"]})
```

```
print("Company Counts:")
```

```
print(company_df["Company"].value_counts())
```

```
data = np.arange(15).reshape((3, 5))
```

```
dframe = pd.DataFrame(
```

```
    data,
```

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```
index=["Rainfall", "Humidity", "Wind"],  
columns=["Bergen", "Oslo", "Trondheim", "Stavanger", "Kristiansand"],  
)  
print("\nOriginal DataFrame:")  
print(dframe)  
stacked = dframe.stack()  
print("\nStacked Data:")  
print(stacked)  
unstacked = stacked.unstack()  
print("\nUnstacked Data:")  
print(unstacked)
```

```
... Company Counts:  
Company  
ABC      3  
XYZ      2  
PQR      1  
Name: count, dtype: int64  
  
Original DataFrame:  
      Bergen  Oslo  Trondheim  Stavanger  Kristiansand  
Rainfall     0     1         2         3         4  
Humidity     5     6         7         8         9  
Wind        10    11        12        13        14  
  
Stacked Data:  
Rainfall  Bergen      0  
          Oslo       1  
          Trondheim  2  
          Stavanger  3  
          Kristiansand 4  
Humidity  Bergen      5  
          Oslo       6  
          Trondheim  7  
          Stavanger  8  
          Kristiansand 9  
Wind      Bergen     10  
          Oslo      11  
          Trondheim 12  
          Stavanger 13  
          Kristiansand 14  
dtype: int64  
  
Unstacked Data:  
      Bergen  Oslo  Trondheim  Stavanger  Kristiansand  
Rainfall     0     1         2         3         4  
Humidity     5     6         7         8         9  
Wind        10    11        12        13        14
```

```
6. series1 = pd.Series([0, 111, 222, 333],
                        index=['zeros', 'ones', 'twos', 'threes'])

series2 = pd.Series([444, 555, 666],
                    index=['fours', 'fives', 'sixs'])

frame = pd.concat([series1, series2],
                  keys=['Number1', 'Number2'])

print("\nHierarchical Series:")
print(frame)

print("\nUnstacked Series:")
print(frame.unstack())
```

```
...
Hierarchical Series:
Number1  zeros      0
         ones     111
         twos     222
         threes    333
Number2  fours     444
         fives     555
         sixs     666
dtype: int64

Unstacked Series:
         fives  fours  ones  sixs  threes  twos  zeros
Number1    NaN   NaN  111.0   NaN   333.0  222.0    0.0
Number2  555.0  444.0   NaN  666.0    NaN   NaN   NaN
```